PROJECT NAME: The "Safe-Night" Transit

Limitations of Safe-Night Transit System:

1. **Basic Routing Logic**
    - Current system uses simple distance calculation
    - Does not consider real traffic or shortest path maps
2. **No Real-Time GPS Tracking**
    - Students cannot track live vehicle location
    - ETA is approximate, not highly accurate
3. **Limited Scalability (Prototype Level)**
    - Designed for small campus use
    - Not optimized for large-scale city deployment
4. **No Authentication System**
    - Student identity verification is basic
    - No secure login (JWT/OAuth not implemented)
5. **No Full Frontend/UI**
    - Focus is mainly on backend logic
    - User interface is not fully developed
6. **Manual Threshold-Based Pooling**
    - Uses fixed detour threshold
    - Not optimized using AI/ML
7. **Single Server Dependency**
    - Centralized system may become a bottleneck
    - Failure of server affects entire system

Future Scope of the Project

1. **Integration with Real Maps**
    - Use OpenStreetMap + OSRM for accurate routing
    - Real-time shortest path calculation
2. **Live Tracking System**
    - GPS-based vehicle tracking
    - Real-time updates to students
3. **AI-Based Smart Pooling**
    - Machine learning to predict demand
    - Better ride clustering and route optimization
4. **Mobile Application Development**
    - Full Android/iOS apps for students and drivers
    - Better user experience
5. **Advanced Security Features**
    - Biometric or RFID-based student verification
    - Emergency contact integration
6. **Scalability Improvements**
    - Cloud deployment (AWS/GCP)
    - Load balancing and microservices architecture
7. **Analytics Dashboard**
    - Heatmaps for demand areas
    - Usage statistics for admin
8. **Integration with Campus Systems**
    - Link with student ID database
    - Automatic verification